

MAT 5173 — Weekly Schedule

Algebra I · Fall 2026 · UT San Antonio

Texts

Primary text

- **Knapp** — Knapp, *Basic Algebra*

Secondary reading

- **Rosebrock** — Rosebrock, *Visual Group Theory*
- **Beardon** — Beardon, *Algebra and Geometry*
- **CLO** — Cox, Little & O'Shea, *Ideals, Varieties, and Algorithms*

Supplementary reading

- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*
- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

Handouts & papers

- **Smith** — Smith, Kahanpaa, Kekalainen & Traves, *An Invitation to Algebraic Geometry*
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*

Schedule

Date		Topic	Reading	Due
Wed 19 Aug		Course launch; the plane and its isometries	Rosebrock Ch. 1	
Mon 24 Aug		Reflections generate everything	Rosebrock Ch. 1	
Wed 26 Aug		From symmetry to the group axioms	Rosebrock Ch. 2; Knapp Ch. IV	
Mon 31 Aug		Cyclic and dihedral groups	Rosebrock Ch. 2	Problem Set 1
Wed 2 Sep	Q&A	Defense 1		
Wed 9 Sep		Permutations and the sign homomorphism	Beardon §1.3-1.4; Knapp Ch. I (permutations)	
Mon 14 Sep		Subgroups; Cayley graphs	Rosebrock Ch. 3	Problem Set 2
Wed 16 Sep	Q&A	Defense 2		
Mon 21 Sep		Cosets and Lagrange	Beardon §12.2-12.3; Knapp Ch. IV	
Wed 23 Sep		Homomorphisms and quotient groups	Rosebrock Ch. 3; Beardon §12.11-12.12	
Mon 28 Sep		Group actions: orbits and stabilizers	Rosebrock Ch. 4; Knapp Ch. IV	

Date		Topic	Reading	Due
Wed 30 Sep	Exam	Midterm Exam 1 — Segment A		
Mon 5 Oct		Orbit-stabilizer and counting	Rosebrock Ch. 4	
Wed 7 Oct		Symmetries of the polyhedra; finite rotation groups in space	Beardon §14.2-14.3	
Wed 14 Oct		Finite groups; frieze and wallpaper patterns	Rosebrock Ch. 7; Beardon §14.4	
Mon 19 Oct		Mobius transformations as a group action on the sphere	Beardon Ch. 13; Arnold–Rogness; Needham Ch. 3 §IV-VI (handout; a copy is on reserve); Mobius Transformations Revealed (2 min)	Problem Set 3
Wed 21 Oct	Q&A	Defense 3		
Mon 26 Oct		Rings and polynomial rings $k[x]$, $k[x,y]$	Knapp §VIII.1; CLO §1.1; Beardon §2.3	
Wed 28 Oct		Integral domains and fields of fractions; polynomials in one variable	Knapp §VIII.2; CLO §1.5	
Mon 2 Nov	Q&A	Ideals; prime and maximal ideals	Knapp §VIII.3; CLO §1.4	Problem Set 4
Wed 4 Nov		Defense 4		
Mon 9 Nov		Affine space and affine varieties $V(S)$	CLO §1.2; Smith Ch. 1 (handout)	
Wed 11 Nov		$I(V)$ and the algebra-geometry dictionary	CLO §1.4	
Mon 16 Nov		The coordinate ring $k[x,y]/I(V)$ — the landing	CLO §1.4; Knapp §VIII.4; Reading guide, Segment B	Problem Set 5
Wed 18 Nov	Q&A	Defense 5		
Mon 23 Nov		Why C and not R ; plane curves — rational, cusp, node	Shafarevich §1.1	
Mon 30 Nov	<i>Review</i>	Review — the exam pool		
Wed 2 Dec	Exam	Midterm Exam 2 — Segment B weighted		